

Machine learning to predict biochar and bio-oil yields from co-pyrolysis of biomass and plastics

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ABSTRACT

Because of high oxygen content, pH and viscosity, pyrolysis bio-oil is of low quality. Upgrading bio-oil can be achieved by co-pyrolysis of biomass with waste plastics, and it is seen as a promising measure for mitigating waste. In this work, machine learning models were developed to predict yields from the co-pyrolysis of biomass and plastics. Classical machine learning and neural network algorithms were trained with datasets, acquired for biochar and bio-oil yields, with cross-validation and hyperparameters. XGBoost predicted biochar yield with an RMSE of 1.77 and R^2 of 0.96, and the dense neural network was able to predict the bio-oil yield with an RMSE of 2.6 and R^2 of 0.96. The SHapley Additive exPlanations analysis technique was used to understand the influence of various parameters on the yields from co-pyrolysis. This study provides valuable insights to understand the co-pyrolysis of biomass and plastics, and it opens the way for further improvements.

1. Introduction

Increasing environmental concerns, demands for energy to serve a growing population, and economic development have driven the utilization of renewable energy sources. Biomass, the only source of renewable carbon, is expected to play a crucial role in sustainable development by enhancing the share of bioenergy in the global energy mix. The International Renewable Energy Agency has estimated the energy potential from biomass to be in the range of 97–147 EJ per year by 2030 [1]. The majority of biomasses are lignocellulosic in nature, originating from agro-residues, energy crops, and forest residues. In considering the potential of biomass, a partial replacement of fossil fuel resources for the production of chemicals and fuel molecules is envisaged. Biomass can be converted into energy and products via biological, chemical, and thermochemical conversion [2].

Thermochemical conversion techniques have gained importance in the production of chemicals and fuel molecules in recent years because of the time, cost, and efficiency concerns associated with biological and chemical conversion. Pyrolysis is the thermochemical conversion process involving the deconstruction of organics into three products: bio-oil, biochar, and gases. Lignocellulosic and algae-based biomasses are promising candidates for the generation of energy and chemicals with low environmental impact, via thermochemical conversion [3]. Pyrolysis has also emerged as an attractive option for waste mitigation [2];

waste is converted into valuable chemicals, while simultaneously achieving significant waste reduction, which otherwise would pervade landfills.

Lignocellulosic biomass is composed mainly of cellulose, hemicellulose, and lignin in varying compositions; these enable it to serve as a platform for many bio-based chemicals and polymers [4]. Their high oxygen content results in significant oxygenated compounds like acids, alcohols, carbonyl compounds, and esters in the bio-oil [5]. The complex composition of bio-oil makes it acidic, polar, and viscous in nature, restricting its direct use [5]. It becomes important to improve the quality of the bio-oils for their use as drop-in fuels and fuel additives [5]. The common upgrading techniques are steam reforming, hydrotreating, and catalytic cracking [6], and its high cost and complex bio-oil composition make processing a challenge [6].

Global plastic production has reached 367 million metric tons, 60% of which ends up in landfills [7]. Pyrolysis of plastics, hydrogen-rich materials, can produce diesel-grade hydrocarbon fuel [8]. In recent years, co-pyrolysis, a single-step conversion technique, has emerged as a simple route to upgrade biomass pyrolysis vapors in the same reactor vessel [6]. The idea of co-pyrolysis was conceived because biomass processing with hydrogen-rich feedstock results in hydrogen transfer to produce an upgraded fuel without requiring a separate hydroprocessing unit [9]. This bio-oil upgrade occurs via the promotion of reactions such as hydrogenation and hydrodeoxygenation.

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Yields of bio-oil and biochar from co-pyrolysis depend on parameters like heating rate, volatile matter, temperature, reaction time, elemental composition, plastic-to-biomass ratio, etc. It is important to understand the key parameters driving the quantity and quality of the bio-oil and biochar produced from co-pyrolysis. In co-pyrolysis, the plastic-to-biomass ratio is an important parameter, having a significant effect on product yield and quality; this was also observed in findings from Dorado et al. [10] and Tang et al. [11]. In a comprehensive review on the co-pyrolysis of biomass and waste plastics by Uzoejinwa et al. [12], the effects of co-pyrolysis parameters were thoroughly examined. The literature on co-pyrolysis of biomass and polymers collated explain that product yields have been experimentally determined and, before arriving at optimum, considerable experimentation is required.

Among all the parameters, the most widely studied are plastic-to-biomass ratio, temperature, and reaction time. It is important to note that other parameters like heating rates and composition of the biomass and plastics are also important. Considering this, the challenge is to comprehensively investigate all the parameters experimentally and assess their effects on product yields.

Supervised machine learning (ML) is a subset of artificial intelligence, a technology that allows computers to emulate human intelligence. ML has been used for complex problems, and to replace conventional statistical methods that require extensive experimenting and tuning [13]. For such problems, ML provides a more accurate and simpler approach [13]. Machine learning provides the computer with the power to learn patterns and complex relationships between parameters without explicitly programming the parameters' relationships [13]. Over the last few years, ML has been proven to capture complex classification and regression problems in multiple fields, including physical sciences, healthcare, and manufacturing [14–16]. The SKLearn and TensorFlow modules provided by SML algorithms can capture complex relationships in chemical properties and the physical conditions of pyrolysis reactions to predict the yields of products [17–21].

Supervised machine learning (ML) has been adopted to provide a more accurate and simpler approach for problems that are too complex for conventional statistical methods [13]. Over the last years, ML algorithms have shown their ability to capture complex classification and regression problems in multiple fields, including physical sciences, healthcare, and manufacturing.

There is little research that addresses the application of ML to understand biomass pyrolysis and predicts the yields. Hough et al. [22] utilized neural networks for the reduction of a detailed kinetic model for lignin pyrolysis. A kinetic model for biomass pyrolysis was developed by Ji et al. [23] using chemical reaction neural networks. ML tools have also been used to predict biochar yield and carbon content [24]. Sunforka et al. [25] developed artificial neural network based models for the evaluation of the kinetic parameters such as apparent activation energy, pre-exponential factor and reaction order for the pyrolysis of biomass. Basically, a correlation between the biomass constituents was constructed with the kinetic parameters. Pyrolysis conditions and biomass characteristics were used as input for developing ML-based models to predict the quality of biochar and pyrolysis gas [26,27]. In a recent study by Wei et al. [28], a random forest and extremely trees approach was utilized to predict mass loss profiles in a thermogravimetric analyzer during co-pyrolysis of biomass and coal. However, to the best of the authors' knowledge, ML has not been used to predict product yields from the co-pyrolysis of biomass and plastics.

Co-pyrolysis of biomass and plastic is sufficiently complex that it warrants the development of mathematical and predictive models to better understand and scale up the process. Co-pyrolysis of biomass and plastic involves numerous simultaneous reactions including heat and mass transfer effects as well. In general, computational fluid dynamics and kinetics-based models have been utilized for the modeling of pyrolysis processes, but these are computationally expensive. As discussed above, ML techniques are promising alternatives proving efficient and effective for modeling these complex systems. The present work adopts a

standard ML approach by mapping inputs to the corresponding outputs without considering the physical and chemical processes occurring during co-pyrolysis. It is important to highlight that the ML-based approach offers the following advantages: (i) no assumptions which may result in inaccurate predictions, (ii) no specific reactor design parameters are needed for modeling, and (iii) low computational cost. The design of a machine learning model for co-pyrolysis problems serves as a foundation for future optimization and integration with advanced simulation tools, which helps highlight the pros and cons of machine learning for co-pyrolysis problems.

In this work, datasets of bio-oil and biochar yields obtained from the literature on co-pyrolysis were collected and used to train the dense neural network (DNN), and XGBoost (XGB) model, without optimization to identify the inherent prediction performance. A grid search was used with cross-validation to identify the hyperparameters for optimization of DNN and XGB. Additionally, SHAP (SHapley Additive exPlanations) clarified the dependence of the features on bio-oil and biochar yield predictions.

2. Methodology

The basis of the experimental data collection was founded upon the literature on co-pyrolysis of mainly lignocellulose biomasses and various plastics. Two sets of experimental data were collected for biochar (94 data points) and bio-oil (96 data points) and used to train, test, and validate the performance of all the models. The data was extracted directly from the result tables in the papers; for some data points, the Grabit MATLAB® package was used to digitize the plots. The datasets used for bio-oil and biochar prediction were tabulated with the sources mentioned and are provided in SM (Supplementary Materials).

The data collected for this study was continuous for all the variables for regression machine learning application without bias or pre-selection. The experimental data collected covered a range of different reactors, lignocellulosic biomasses and polymers. However, the models were designed to be reactor, biomass, and plastic agnostic. The collected data were chosen to reflect the chemical interactions and physical processes, based on the intrinsic properties of the materials and reaction conditions. The variables chosen in this study were the characteristics of the biomass and plastics, reaction conditions, and plastic percent in the sample (w% polymer). Characteristics of the biomass used were the proximate and ultimate analyses, whereas the ultimate analysis was used for plastics. The ultimate analysis included carbon, hydrogen, oxygen and nitrogen content by weight, while the proximate analysis included ash, fixed carbon, volatile matter, and moisture by weight. The reaction conditions considered were C degrees (pyrolysis temperature, in °C), h (reaction time, in h), C/min (heating rate, in °C min⁻¹).

The proximate analysis of polymers was not considered as an input as most plastics such as polypropylene, polystyrene and polyethylene present in the waste comprise high volatile matter (~99%) [29]. It can be noted that the particle size was also not included as an input for co-pyrolysis experiments. The rationale behind this is the homogeneous mixing of the samples investigated for co-pyrolysis, generally achieved by grinding, which ensured that the sample for an experiment has a uniform particle size (1–2 mm) across all the studies. As the value of these parameters (proximate analysis of the plastics and the particle size) would be the same across all the data points considered as inputs, the predictions of the yields of products would not be affected.

The concept of stratified cross-validation was implemented as part of the data pretreatment to ensure that the validation test sample had the same distribution as the training data. The stratified sampling technique was used with a Pearson correlation analysis. Among the experimental data, the most influential linear correlation factor in both biochar and bio-oil was found to be the plastic-to-biomass ratio. The stratification of the experimental data was performed by designing five-to-six ranged bins for the distribution of the training data and final validation data. It should be noted that each bin was randomly split into training data and

final validation data before finally being merged to fit and test the models; this prevented the initial random split from the allocation of test data out of the training range, ensuring good distribution of the data used to train the models. Data feature scaling and normalization were conducted because some machine learning algorithms do not distribute the data well with widely disparate values of the features, which can result in excess importance on some features [13]. The null values were a couple of points that have been replaced with the mean or mode as appropriate.

The machine-learning algorithms of the SKLearn and TensorFlow modules captured the complex relationship of the chemical composition, the physical conditions of pyrolysis reactions to predict product yields, and other applications [17–21]. The modules' models were used with the cross-validation with KFold ($N = 10$); this technique is commonly used to prevent the overfitting of the training data, and it helps the algorithm to capture the actual parameters to retrain the model for final validation [13]. The structure of the procedures utilized in this study is presented in Fig. 1.

The models of interest, DNN and XGB, with cross-validation and were tested with final validation data. It can be noted that the data points in the validation data were not utilized for training either of the chosen models. The performance of the classic machine learning model and the neural network model was investigated. XGB, a random forest-based model, and DNN models were further optimized with the grid search to obtain the hyperparameters and retrain the models for final validation, as shown in Fig. 1. As per the cheat sheet for SKLearn algorithms used for choosing the right estimator, more than 50 data points are sufficient to develop a machine learning estimator. So, the dataset collected can be foreseen to predict the products' yields reliably. Because of the small size of the dataset (~ 100 points) for biochar and bio-oil yields, the initial split of the data was determined to be 90% training and 10% final validation. With this training procedure, the probability of overfitting is minimized because cross-validation and the final validation data help assess the fitting of the models [30]. The final validation data has been hidden from the model until the very end to test the performance of the models. Furthermore, the SHAP analysis helps verify whether the model has captured the relationship between the

features and target outputs.

Using the grid search method from the SKLearn module, hyperparameters were found by defining a search space for the parameters for XGB and DNN models. The hyperparameters investigated for XGB were the number of estimators, maximum features, maximum depth, minimum samples split, minimum sample leaf, and bootstrap. For the DNN model, the hyperparameters investigated were epochs, learning rate, loss function, dropout rate, batch size, and the number of neurons. After the hyperparameters were obtained, the models were tested on different iterations of the initial split to further randomize the process.

The performance of the models was measured using three validation factors, the root mean square error (RMSE), the regression coefficient (R^2), and the mean absolute error (MAE). These performance measures assessed the effectiveness of the models developed and clarified the offset in the magnitude of the actual yield of biochar and bio-oil from the yields predicted by the models [13]. RMSE represents the square root of the average of the difference between actual data and predictions squared (Equation (1)) in which y_i , $\hat{y}$, and $\bar{y}$ represent the actual instance, predicted instance, and mean value of y , respectively. MAE represents the average of the absolute value of the difference between actual data and predictions (Equation (2)). Both RMSE and MAE quantify the average of the error associated with the prediction of the model. However, the RMSE penalizes outliers and large prediction errors in the regression of the final validation dataset, and RMSE is considered a standard for evaluating regression models [24,26,27]. The lower values of RMSE and MAE are preferred as they indicate a low error in the predictions associated with the developed models. The R-square is a metric to represent the proportion of the variance and the linearity between the predictions and actual data, represented (Equation (3)). A high R-square value indicates that there was no high variance in the prediction of the yields using the model, which is preferable. Nonetheless, RMSE is a better indicator of the accuracy of the models predicting char and liquid yields.

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y})^2} \quad (1)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}| \quad (2)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y})^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (3)$$

For a detailed analysis of the features used for training, the SHAP (Shapley additive explanations) module was used to understand the output and impact of training features. SHAP values are commonly used tools, originating from a cooperative game theory, which explains the output of machine learning models by translating the optimal credit distribution with value-based (local) explanations. SHAP analysis was carried out for both biochar and bio-oil in the DNN and XGB models, resulting in the development of a bar plot that represents mean absolute SHAP values, impact values on the model prediction for each training feature, and a beeswarm plot representing the impact on the model output of high to low values for each training feature. These machine learning tools aid in investigating non-linear interactions of the features in the co-pyrolysis of biomass and plastics.

SHAP has different core explainers suited for different well-known models. For instance, the Tree Explainer is used in SHAP analysis for the XGBoost model since it's a tree-based model. SHAP provides a localized interpretation of each feature's contribution during the training and fitting of the data. An important XGBoost attribute is the max feature which allows the model to sample features and fit data systematically. However, the DNN model considers all features involved at once and assigns appropriate weights to each feature. As a result the SHAP analysis results are different for the DNN.

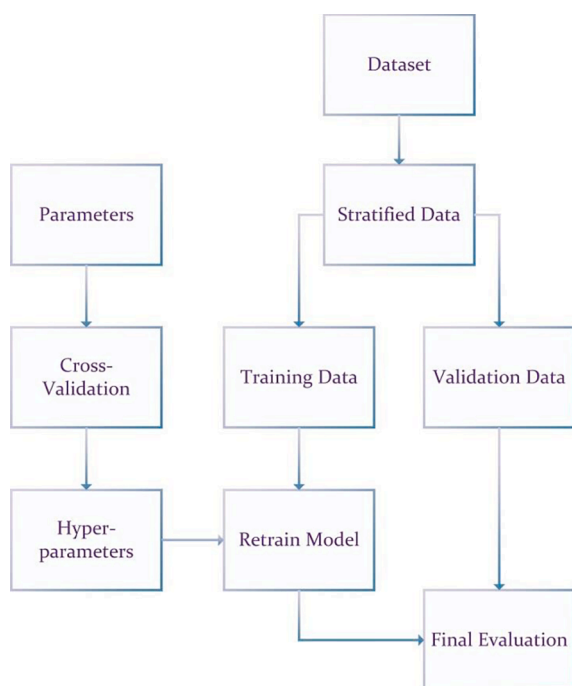


Fig. 1. Training flow of machine learning models and optimization via cross-validation and hyper-parameters.

3. Results and discussion

3.1. Significance of proximate analyses and data stratification

The features used for training were tested with and without proximate analysis of biomass as input. Previous research showed that the ability of the model to predict the quantity of biochar was not significantly affected when only ultimate analysis, or only proximate analysis, or both, were considered for training the models [24]. The biochar Pearson's correlation heat map indicated that char yield is strongly correlated with both features of ultimate and proximate analyses, as seen in Fig. 2.

The Pearson's correlations for the carbon and oxygen content in the biomass were -0.35 and 0.37 , respectively, whereas the correlations for the carbon and oxygen content in the polymer were -0.24 and 0.42 , respectively, in biochar data. Finally, biomass volatile matter and fixed carbon were found to have correlations of -0.46 and 0.39 , respectively. It can be inferred that the biomass volatile matter and fixed carbon correlated significantly with the biochar yield in the co-pyrolysis. The detailed effect of fixed carbon and volatile matter of biomass on the prediction of biochar yields was performed using SHAP feature analysis.

For bio-oil, however, the magnitudes of Pearson's correlations for C, H, N, and O content were low (<0.2) (Fig. 3). This resulted in an inaccurate prediction of the bio-oil yields during co-pyrolysis. The XGB model, without proximate analysis data, had RMSE, R^2 , and MAE values of 5.25, 0.82, and 3.71, respectively. While training the model, the inclusion of proximate analysis data enhanced the prediction ability of the model. RMSE and MAE significantly decreased to 3.11 and 2.6, and the R^2 value increased to 0.94. With the inclusion of the proximate analysis of the biomass in training the model, it became responsive toward the volatile and fixed carbon fraction present in the biomass, which was not

considered previously in the prediction of bio-oil yield. The detailed effect of fixed carbon and volatile matter of biomass on the prediction of bio-oil was explained using SHAP feature analysis.

The stratification of the data significantly enhanced the predicting ability of the models for yields of co-pyrolysis products. Although it was randomized, the distribution of data without stratification was uneven and this was attributed to the small dataset size. The split of the initial training data and final validation data was set at 90% training /10% final validation data from the original biochar and bio-oil dataset. Other combinations of data splitting were also investigated, such as 80% training/20% final validation data, and 70% training data/30% final validation data. From the yield predictions, it appeared that the models struggled to learn from the data due to a reduction in the number of points used by the models to train. With data stratification, it was possible to distribute the final validation data evenly and randomly over the training range.

3.2. Analysis of training features and their impact on model prediction

Using SHAP values, the training features for model yield prediction were analyzed. Figs. 4 and 5 show the mean absolute value of each training feature for every model and their impact on the model's predictions (training feature values from high to low).

The DNN models assigned significant SHAP values to most of the training features, while the XGB models credited SHAP values to the most important contributing features: Mean values of the features that trained the DNN model were high compared to each other. For the XGB, mean values were high for only a few features and low for the remainder. This analysis helps in understanding how the models learn the relationship between the training features and their significance in yield prediction. XGB and SHAP analysis have been applied to road

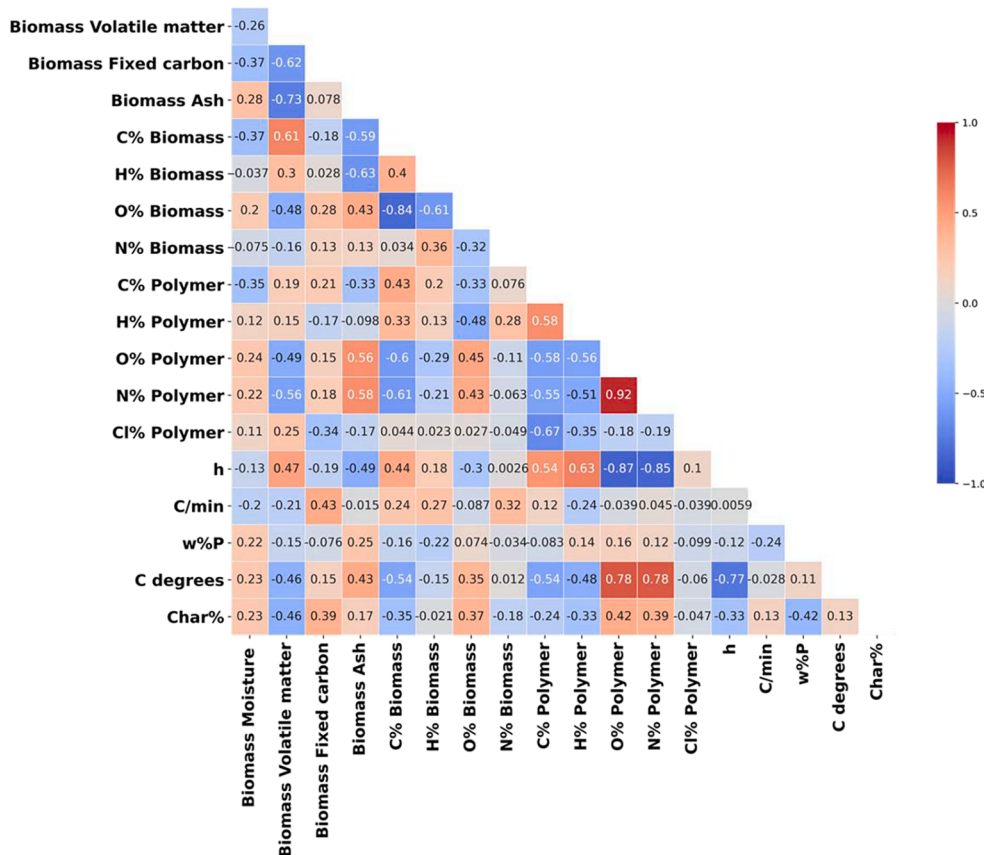


Fig. 2. Heat map of Pearson's correlations of parameters for biochar used in this work, where h, C/min, C degrees, and w%P represent reaction time (in min), heating rate ($^{\circ}\text{C}/\text{min}$), pyrolysis temperature ($^{\circ}\text{C}$) and plastic weight percent in the sample, respectively.

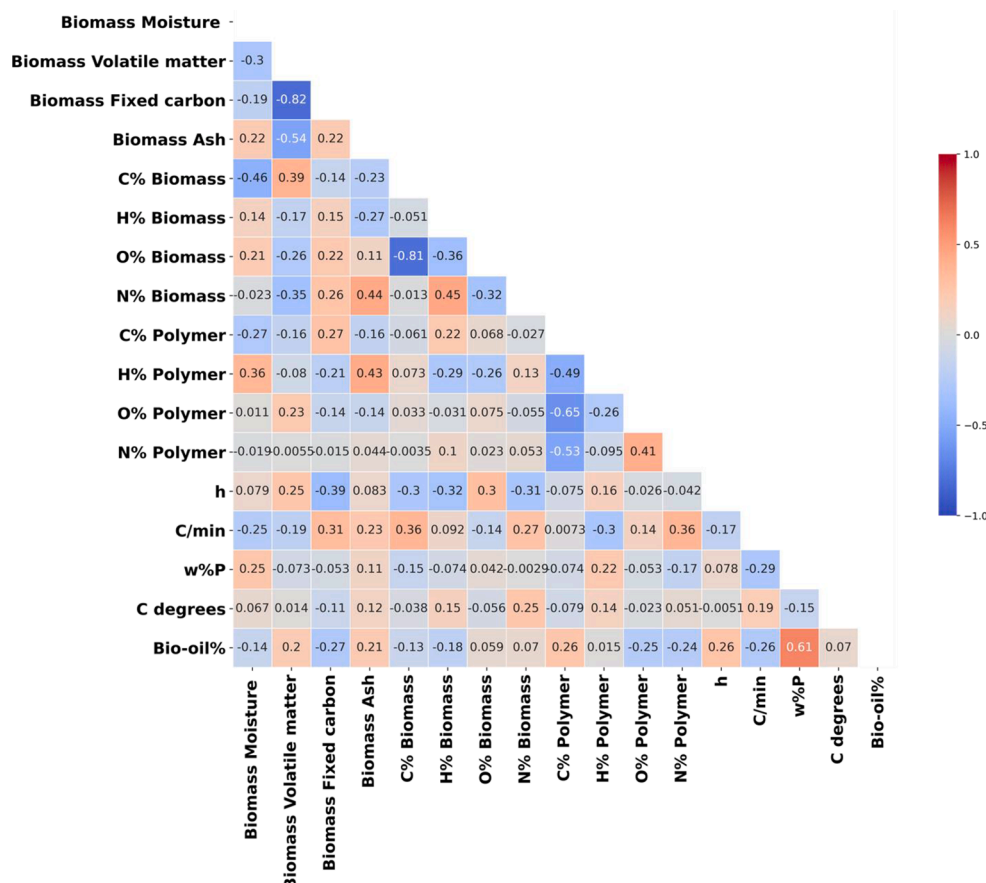


Fig. 3. Heat map of Pearson's correlations of parameters for bio-oil used in this work, where h, C/min, C degrees, and w%P represent reaction time (in min), heating rate ($^{\circ}\text{C}/\text{min}$), pyrolysis temperature ($^{\circ}\text{C}$) and plastic weight percent in the sample, respectively.

safety analysis, healthcare, and solid-state chemistry to understand the impact of features on predictions [31–33].

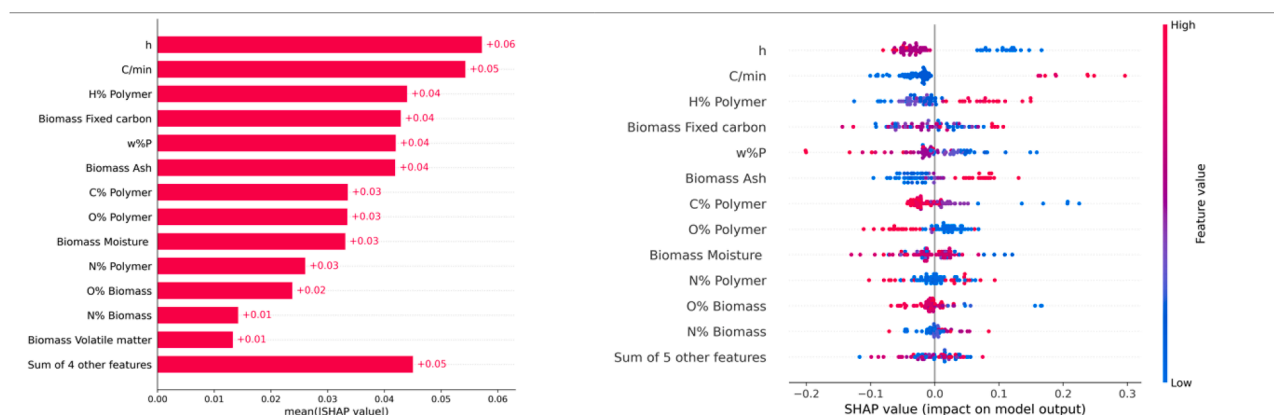
Fig. 4 (a) and (b) show mean values of the features used to train biochar and bio-oil data, respectively, to predict yields using XGB. For XGB model SHAP analysis, the top six training features appeared to have a significant impact on yield prediction, while other features had a minimal impact on predictions. The weight percent of polymers in the mixture was identified to have a large contribution to the accurate prediction of biochar and bio-oil models with 3.88 and 6.47 mean absolute SHAP values, respectively. This is also reflected in the impact of w%P in the prediction: high value resulted in low and high biochar, and bio-oil, respectively; more plastic in the feedstock resulted in more liquid yield and less char yield from the pyrolysis [30]. Fixed carbon in biomass controls char formation; more fixed carbon results in more char formation during biomass pyrolysis, across all pyrolysis temperatures [31]. SHAP analysis determined that biomass fixed carbon was the second most dominant feature in predicting biochar yield, with a mean value of 2.47. The impact of fixed carbon suggested the same conclusion. Biomass with more fixed carbon resulted in more solid residue from the co-pyrolysis [6,34,35].

The temperature of the reaction was found to have significant impact on the yields predictions for both SHAP analysis of features for biochar and bio-oil XGB models. Beyond the optimum pyrolysis temperature, an increase in temperature reduced bio-oil yield. This can be attributed to enhanced cracking reactions at higher temperatures where the condensable fraction responsible for the formation of bio-oil cracks to non-condensable gases. However, beyond a specific pyrolysis temperature, biochar yields were not affected. This can be attributed to the fact that after the complete devolatilization of the biomass, the increase in pyrolysis temperature did not deconstruct the fixed carbon under pyrolysis

conditions. It can be noted that under pyrolysis conditions, the maximum possible biochar yield is the fixed carbon content in the biomass. The biochar yield decreases as the temperature increased. The mean SHAP value for temperature in the bio-oil and biochar XGB model was 2.55 and 1.82, respectively. The impact of temperature on prediction was also correctly captured; it correlated well with the actual physical phenomenon. Biomass volatile matter and moisture had mean SHAP values of 1.65 and 0.84, respectively for the biochar XGB model. The beeswarm plot suggested that low and high volatile matter in the biomass translated to high and low biochar yields, respectively; this corroborated well with the biomass pyrolysis. The impact of moisture on bio-oil yield was positive, i.e., higher moisture content resulted in a high amount of char; this was also evidenced by Bridgwater and coworkers [36], as pyrolysis of pre-treated trommel fines at 500°C , with moisture 2.7%, 5%, and 10% resulting in the char yields of 13.7%, 14.6%, and 17.4%, respectively.

Following is a discussion on the impact of various features on the prediction of bio-oil yields via the XGB model. The SHAP mean value of biomass moisture was 1.82. High values of moisture suggested a high value of bio-oil, as captured by the beeswarm plot. This is partially correct, as bio-oil obtained from pyrolysis also contains a significant amount of water, and more moisture results in an overall higher liquid yield [37]. However, data on co-pyrolysis in the literature does not distinguish between the energy-dense bio-oil and water, separately. All the studies report liquid yields, which this study considered as bio-oil yield. It was interesting that the carbon content of the plastics was also found to play an important role in predicting bio-oil and biochar yields using the XGB model. The mean SHAP value for the bio-oil and biochar XGB models were 1.36 and 1.13, respectively. The higher carbon content in plastics resulted in low char and high bio-oil yields. This

(a) SHAP Analysis Biochar DNN



(b) SHAP Analysis Bio-oil DNN

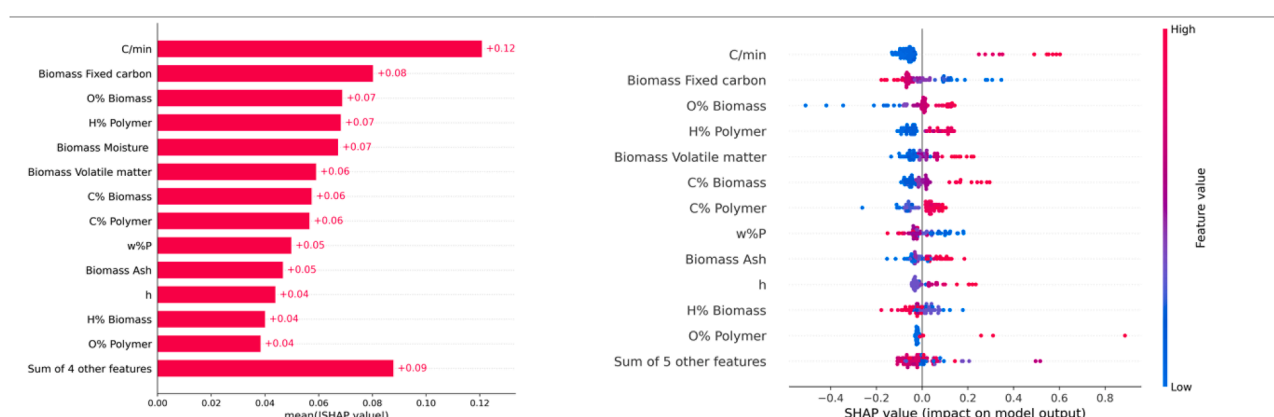


Fig. 4. SHAP analysis diagrams of bar plot of mean absolute SHAP values (left) and beeswarm plot of SHAP values (right) of (a) SHAP analysis of biochar XGB model (b) SHAP analysis of bio-oil XGB model.

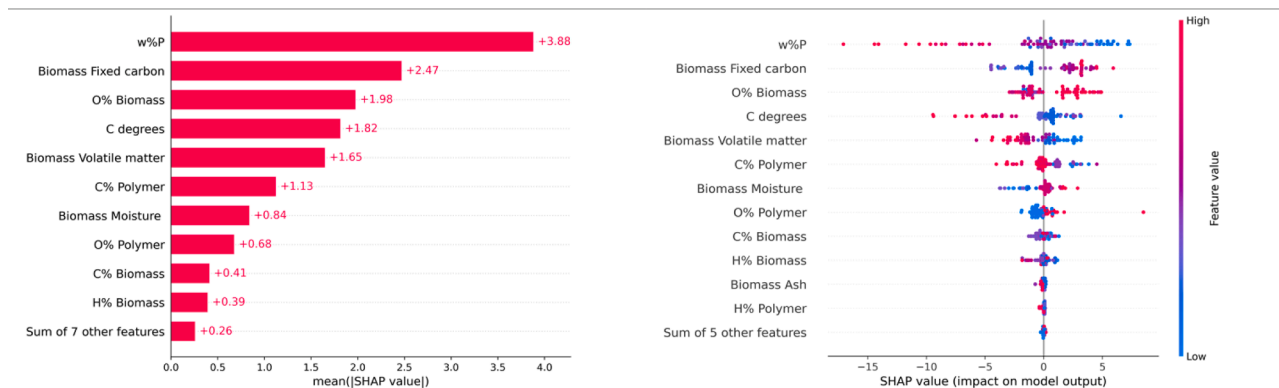
is well reported in the literature on co-pyrolysis of biomass and plastics [6]. The carbon content of the biomass was also found to have a noticeable impact on the prediction of bio-oil yield, using the XGB model, and the corresponding SHAP mean value was 1.3. The high carbon content suggested a lower bio-oil yield, attributable to the volatile carbon remaining in the solid residue and not being released in the vapors, which would result in the formation of bio-oil.

Fig. 5 (a) and (b) depict the mean SHAP values of the features used to train the DNN models and their impact on the prediction of the yields of biochar and bio-oil, respectively. The mean SHAP values are lower than the XGB model. SHAP analysis of DNN models reflected the normalized outputs from the DNN training process. Time (h) was the greatest contributor (0.6) to biochar yield prediction and its impact on yield was correctly captured as longer reaction times reduced the char yield and enhanced gas formation. As the reaction time increased, char yield decreased, because of the cracking of the biomass/plastics mixtures [38]. As seen in the XGB model, biomass fixed carbon, ash, and the plastic-to-biomass ratio also contributed significantly to the prediction of biochar yields. This was observed from the SHAP mean values and impact plots. High fixed carbon and ash in the biomass resulted in high char formation during pyrolysis, whereas a high plastic amount resulted in reduced amount of biochar because plastics leave little or no residue during pyrolysis. Biomass volatile matter was also found to have a role in the prediction, with a mean SHAP value of 0.1. Biomass with high volatile matter resulted in low biochar values; this was also captured by the impact plots.

Fig. 5 (b) shows that the heating rate (C/min) was the major contributor to the prediction of bio-oil yields, with a mean SHAP value of 0.12. The impact plot also demonstrated that higher heating rates resulted in a high amount of bio-oil, and vice-versa. As observed in the XGB, biomass characteristics such as fixed carbon, moisture, and volatile matter controlled the prediction of bio-oil yields. The plastic-to-biomass ratio contributed to the prediction of bio-oil yields from both the XGB and DNN models. Experiments also showed that the addition of polymers to the biomass enhanced bio-oil yields, owing to the effects of interactions [34]. The carbon content of both biomass and plastics was also found to contribute significantly to the prediction of bio-oil yields, using both models. The developed models captured the effects of important features on the yields of biochar and bio-oil.

Overall, SHAP analysis of the XGB and DNN model captured the effect of features on the prediction of bio-oil and biochar yields, following the physical phenomena observed during pyrolysis of biomass and polymers. Figs. 4 and 5 show that many parameters affected the prediction of pyrolysis product yields; the w%P, %C of biomass and plastic, temperature, volatile matter, moisture, and fixed carbon emerged to control the process and accurately predict product yields. The role of these parameters can be physically understood in the co-pyrolysis process. This analysis substantiates the performance of individual models discussed in the previous sections. Figs. 6 and 7 unravel the contribution of all the input features in the prediction, and it is important to note that explaining the role of each feature in the prediction is difficult, owing to the complexity of the co-pyrolysis process.

(a) SHAP Analysis Biochar XGB



(b) SHAP Analysis Bio-oil XGB

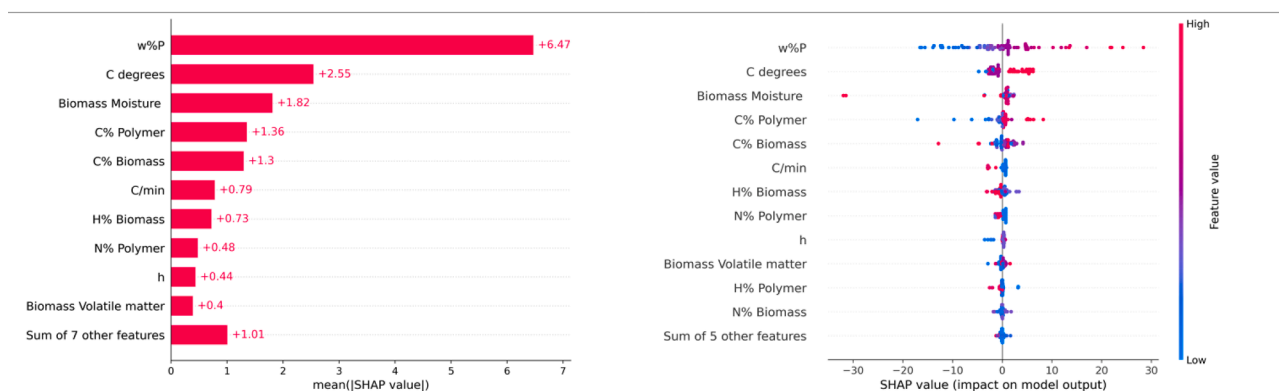


Fig. 5. SHAP analysis diagrams of bar plot of mean absolute SHAP values (left) and beeswarm plot of SHAP values (right) of (a) SHAP analysis of biochar DNN model (b) SHAP analysis of bio-oil DNN model.

Also, the model interprets some of the features in a manner that may not have physical significance, as found in some of the features in the SHAP analysis.

3.3. Evaluation of the performance of various models

The regression algorithms were evaluated to predict the bio-oil and biochar yields during the co-pyrolysis of biomass with polymers. The models were trained and tested with cross-validation (KFold = 10) and final validation data to obtain the performance metrics for the models, RMSE, R^2 and MAE. The neural network model was developed with simple architecture: one normalization layer, three dense neural layers, two dropout neural layers, and one output neural layer for both biochar and bio-oil models. This simple architecture achieved a cross-validation RMSE score of 5.41 ± 4.69 for biochar final validation data prediction, and 6.62 ± 5.24 for bio-oil final validation data prediction with no optimization, for the hyperparameter grid search. The parameters for training the default model for the DNN were randomly chosen as follows: epochs = 300, batch size = 32, loss = mean squared error, number of neurons = 32, and dropout rate = 0.2 for both biochar and bio-oil.

XGB and DNN models were further optimized for the prediction of biochar and bio-oil yield. The parameters optimized for the XGBoost model were as follows: the number of estimators, maximum features, maximum depth, bootstrap, minimum samples split, and minimum samples leaf. The parameters optimized for the DNN model were epochs, batch size, number of neurons, and dropout rate. A random search with cross-validation determined the magnitude of each hyperparameter,

followed by a grid search to find the best hyperparameters for both biochar and bio-oil models. The final validation evaluation was performed on the retrained models, with validation data and plotted the prediction yield vs. actual yield, as shown in Fig. 6 (a)–(d).

The performance of each model was evaluated, and biochar predictions with XGB showed the best accuracy, with an RMSE of 1.77, after optimizing the hyperparameters. The hyperparameters for the biochar XGB model were as follows: the number of estimators, 829; maximum features, auto; maximum depth, 15; bootstrap, true; minimum samples split, 2; and minimum samples leaf, 1. For the XGBoost bio-oil model, the best performance after optimization of hyperparameters was achieved with an RMSE of 3.11. The hyperparameters for the bio-oil XGBoost model were found to be the number of estimators, 294; maximum features, auto; maximum depth, 5; bootstrap is true; the minimum number of samples split was two and the minimum sample leaf was one.

The DNN model with optimized hyperparameters predicted bio-oil data more accurately, with an RMSE of 2.60, compared to the XGB model (RMSE = 3.11). The hyperparameters for the bio-oil DNN model were as follows: epochs = 550, batch size = 32, loss = mean squared error, number of neurons = 32, and dropout rate = 0.1. On the other hand, the biochar DNN model with optimized hyperparameters predicted the yields with an RMSE of 3.26, compared to 1.77 for the XGB. The hyperparameters for the biochar DNN model were: epochs = 450, batch size = 32, loss = mean squared error, number of neurons = 64, and dropout rate = 0.1. Loss function plots are presented in Fig. 6 (e) and (f). This clearly showed how DNN was learning the relationship between the

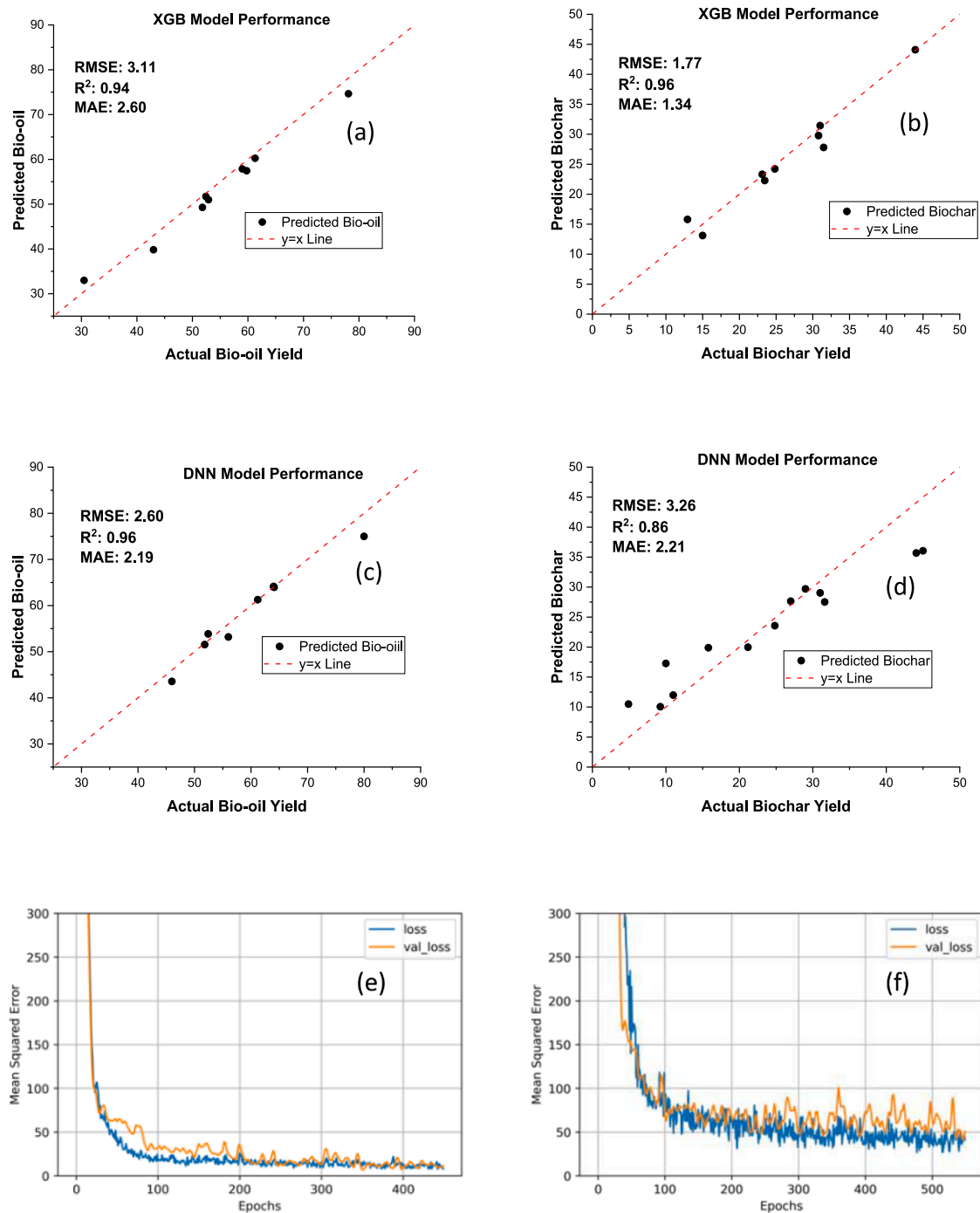


Fig. 6. (a). Plot of predicted bio-oil vs. actual bio-oil yield for XGB model, (b) Plot of predicted biochar vs actual biochar yield for XGB model, (c) Plot of predicted bio-oil vs actual bio-oil yield for DNN model, (d) Plot of predicted biochar vs actual biochar yield for DNN model, (e) Plot of loss and validation loss function for char DNN training and (f) Plot of loss and validation loss function for liquid DNN training.

features with every epoch number. The small size of the biochar and bio-oil datasets were sufficient to train the models to predict accurately without overfitting. The number of neurons for bio-oil hyperparameters shows a good agreement with the rule of thumb presented in Huang [39] for simple neural network architecture. The number of neurons that has the learning capability and capacity to minimize error is approximated as $2\sqrt{(m+2)N}$ where m is number of output and N is the number of samples [39]. This justifies the magnitude of the number of neurons utilized for DNN models.

The models were trained with random functions in the initial split and cross-validation, but to further randomize model training and

validation, the initial splits were randomly performed on different iterations. The results of all the iterations for each model were compiled and plotted together, as shown in Fig. 7 (a)–(d). The overall predictive performance of the models decreased slightly as the RMSE increased. From these plots, the models predicted the co-pyrolysis product yields in the regime where the plastic-to-biomass ratio was not zero and one. Yield predictions deviated from the actual points in pure biomasses and plastic pyrolysis processes because the data points of bio-oil and bio-char yields with pure biomass and plastic pyrolysis were highly limited compared to the number of co-pyrolysis data points. Notably, the models in this study were developed to predict bio-oil and biochar yields

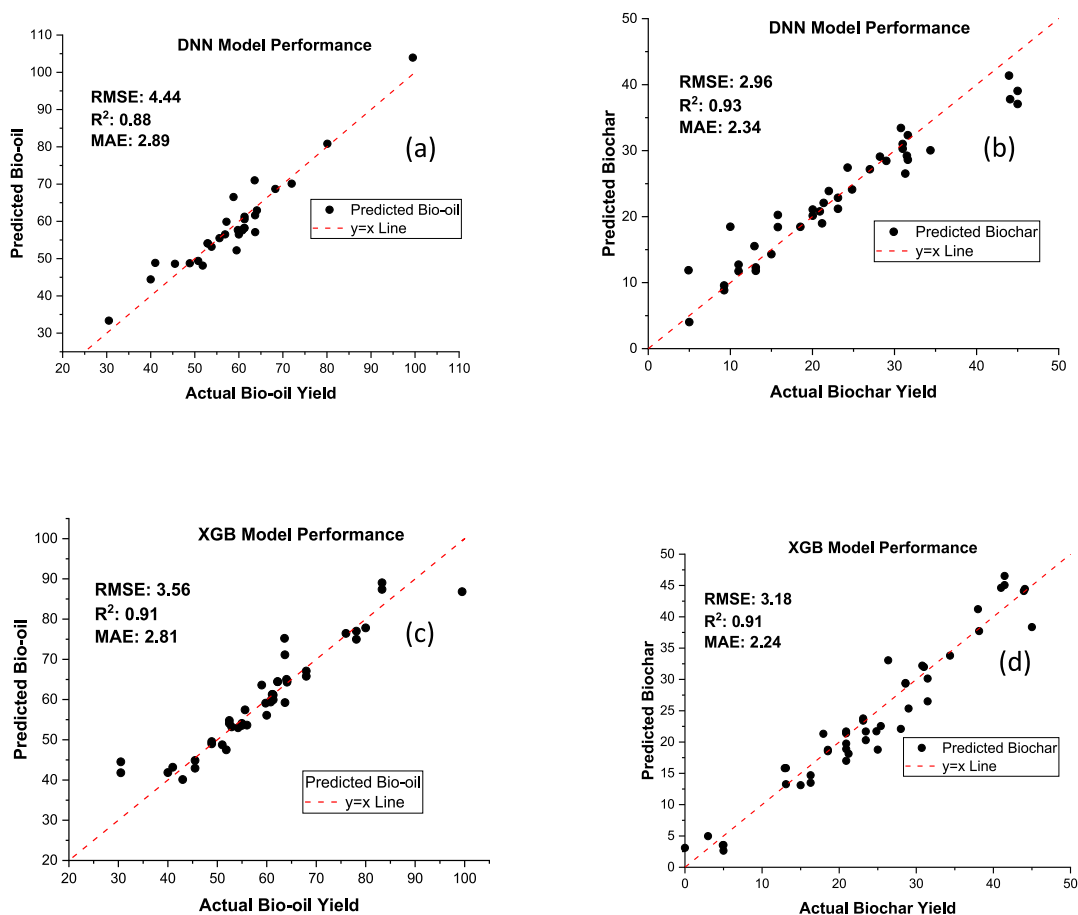


Fig. 7. (a). Plot of combined iterations of predicted bio-oil vs actual bio-oil yield for DNN model, (b) Plot of combined iterations of predicted biochar vs actual biochar yield for DNN model, (c) Plot of combined iterations of predicted bio-oil vs actual bio-oil yield for XGB model and (d) Plot of combined iterations of predicted biochar vs actual biochar yield for XGB model.

from the co-pyrolysis of biomass and plastics.

The performance obtained is comparable to other machine learning models utilized for pyrolysis of biomass alone. For the work by Zhu et al [16], the RMSE associated with the biochar prediction is 3.40 and R^2 of 0.85; whereas, the RMSE for the biochar model in this work is 3.18 and R^2 of 0.91. The bio-oil model published by Tang et al [18] had an RMSE of 2.13 and R^2 of 0.92; whereas, the RMSE of bio-oil model in this study is 2.83 and R^2 of 0.95. The performance of machine learning models developed in this study for both biomass and plastic co-pyrolysis show similar prediction capability and limitations as models developed only for biomass pyrolysis.

From this discussion, it is clear that multiple factors control bio-oil and biochar yields. The factors revealed by the machine learning models developed in this study agree with experimental data present in the literature. SHAP analysis was used to determine an impact of a feature on the prediction of yields. The key factors affecting products' yields were the weight percentage of plastic, biomass fixed carbon, biomass volatile matter, heating rates and pyrolysis temperatures. The weight percentage of plastic directly controls the yield of bio-oil and biochar as plastics are composed of volatiles. Biomass fixed carbon and volatile matter determine the amount of biochar and bio-oil, respectively, from the pyrolysis. Low heating rates enhance the formation of biochar and gases, whereas higher heating rates facilitate the formation of bio-oil. Finally, low temperatures do not provide enough thermal energy to the feedstock which leaves behind a high amount of volatile matter-rich biochar, whereas high temperatures facilitate cracking which produces less biochar and more gases.

4. Conclusions

This study successfully predicted biochar and bio-oil yields from the co-pyrolysis of biomass and plastics using feedstock characteristics, reaction conditions and plastic amounts in the feed with machine learning models. XGBoost and dense neural network models were found to be excellent at predicting yields of biomass and plastics co-pyrolysis. The optimization of hyperparameters also improved predictions significantly, and the inclusion of proximate analysis in the features improved predictions for bio-oil. SHAP analysis has helped to understand the co-pyrolysis of biomass and plastics in light of the effect of parameters driving the formation of bio-oil and biochar. The studies in the literature that used machine learning techniques do not compare the results with traditional models. The results from this study offer a methodology to identify the effect of features on the prediction of bio-oil and biochar yields from the co-pyrolysis of biomass and plastics. The utilization of SHAP with XGB and DNN can be a promising method for predicting the makeup of bio-oil and biochar after the inclusion of biochemical composition in the features. The results will also aid in predicting gas yields from the co-pyrolysis of biomass and plastics.

CCRediT authorship contribution statement

Aessa Alabdrabalnabi: Conceptualization, Methodology, Formal analysis, Investigation, Visualization, Data curation, Software, Validation, Writing - original draft, Writing - review & editing. **Ribhu Gautam:** Conceptualization, Formal analysis, Investigation, Visualization, Data curation, Methodology, Writing - original draft, Writing - review &

editing. **S. Mani Sarathy**: Conceptualization, Supervision, Funding acquisition, Writing - original draft, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used in this study is provided in the [Supplementary data](#)

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.fuel.2022.125303>.

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